

3 章 関数とグラフ

1 2 次関数

BASIC

■143 (1)

$$f(x) = 3x + 1$$

$$f(2) = 6 + 1 = 7$$

$$f(-2) = -6 + 1 = -5$$

$$f(a+2) = 3(a+2) + 1 = 3a + 7$$

$$f(a-2) = 3(a-2) + 1 = 3a - 5$$

■144 (1)

$$y = 2x + 1 \quad (-1 \leq x \leq 2)$$

$$\text{値域: } -1 \leq y \leq 5$$

■144 (2)

$$y = -2x + 4 \quad (-1 \leq x \leq 1)$$

$$\text{値域: } 2 \leq y \leq 6$$

■145 (1)

$$y = x^2 - 1$$

$$\text{軸: } x = 0$$

$$\text{頂点: } (0, -1)$$

■145 (2)

$$y = -(x-1)^2$$

$$\text{軸: } x = 1$$

$$\text{頂点: } (1, 0)$$

■145 (3)

$$y = 3(x+1)^2 - 2$$

$$\text{軸: } x = -1$$

$$\text{頂点: } (-1, -2)$$

■145 (4)

$$y = -\frac{1}{2}(x-2)^2 + 2$$

$$\text{軸: } x = 2$$

$$\text{頂点: } (2, 2)$$

■146 (1)

$$x \rightarrow x-3 \text{ でおきかえて}$$

$$y = 2(x-3)^2$$

■146 (2)

$$x \rightarrow x-1, y \rightarrow y-2 \text{ でおきかえて}$$

$$y = 2(x-1)^2 + 2$$

■146 (3)

$$y = 2(x+2)^2 - 1$$

■147 (1)

$$y = (x-1)^2$$

$$\text{軸: } x = 1$$

$$\text{頂点: } (1, 0)$$

■147 (2)

$$y = 2x^2 - 4x + 3$$

$$= 2(x-1)^2 + 1$$

$$\text{軸: } x = 1$$

$$\text{頂点: } (1, 1)$$

■147 (3)

$$y = -x^2 + 4x - 3$$

$$= -(x-2)^2 + 1$$

$$\text{軸: } x = 2$$

$$\text{頂点: } (2, 1)$$

■147 (4)

$$y = x^2 - x - 2$$

$$= \left(x - \frac{1}{2}\right)^2 - \frac{9}{4}$$

$$\text{軸: } x = \frac{1}{2}$$

$$\text{頂点: } \left(\frac{1}{2}, -\frac{9}{4}\right)$$

■147 (5)

$$y = -\frac{1}{2}x^2 + x$$

$$= -\frac{1}{2}(x-1)^2 + \frac{1}{2}$$

$$\text{軸: } x = 1$$

$$\text{頂点: } \left(1, \frac{1}{2}\right)$$

■147 (6)

$$\begin{aligned}
 y &= -2x^2 - 3x + 2 \\
 &= -2\left(x + \frac{3}{4}\right)^2 + \frac{25}{8} \\
 \text{軸} : x &= -\frac{3}{4} \\
 \text{頂点} : &\left(-\frac{3}{4}, \frac{25}{8}\right)
 \end{aligned}$$

■148 (1)

$$\begin{aligned}
 y &= a(x-3)^2 + 5 \\
 -4 &= a(-3)^2 + 5 \\
 a &= -1 \\
 y &= -(x-3)^2 + 5
 \end{aligned}$$

■148 (2)

$$\begin{aligned}
 y &= ax^2 + q \\
 (-1, 1), (2, -5) &\text{ を通るので} \\
 \begin{cases} 1 = a + q \\ -5 = 4a + q \end{cases} &\text{ より } \begin{cases} a = -2 \\ q = 3 \end{cases} \\
 y &= -2x^2 + 3
 \end{aligned}$$

■148 (3)

$$\begin{aligned}
 y &= a(x-2)^2 + q \\
 (0, 5), (3, 2) &\text{ を通るので} \\
 \begin{cases} 5 = 4a + q \\ 2 = a + q \end{cases} &\text{ より } \begin{cases} a = 1 \\ q = 1 \end{cases} \\
 y &= (x-2)^2 + 1
 \end{aligned}$$

■149 (1)

$$\begin{aligned}
 y &= ax^2 + bx + c \text{ とすると} \\
 \begin{cases} 0 = a + b + c \\ 3 = c \\ -1 = 4a + 2b + c \end{cases} &\text{ より } \begin{cases} a = 1 \\ b = -4 \\ c = 3 \end{cases} \\
 y &= x^2 - 4x + 3 \\
 &= (x-2)^2 - 1 \\
 \text{軸} : x &= 2 \\
 \text{頂点} : &(2, -1)
 \end{aligned}$$

■149 (2)

$$\begin{aligned}
 y &= a(x-3)(x+2) \text{ とすると} \\
 6 &= a \times (-3) \times 2 \\
 a &= -1 \\
 y &= -(x-3)(x+2) \\
 &= -x^2 + x + 6 \\
 &= -\left(x - \frac{1}{2}\right)^2 + \frac{25}{4} \\
 \text{軸} : x &= \frac{1}{2} \\
 \text{頂点} : &\left(\frac{1}{2}, \frac{25}{4}\right)
 \end{aligned}$$

■150 (1)

$$\begin{aligned}
 y &= x^2 - 4x + 8 \\
 &= (x-2)^2 + 4 \\
 x = 2 &\text{ で最小値 } 4, \text{ 最大値なし}
 \end{aligned}$$

■150 (2)

$$\begin{aligned}
 y &= -x^2 + 6x - 1 \\
 &= -(x-3)^2 + 8 \\
 x = 3 &\text{ で最大値 } 8, \text{ 最小値なし}
 \end{aligned}$$

■150 (3)

$$\begin{aligned}
 y &= \frac{1}{2}x^2 - 2x \\
 &= \frac{1}{2}(x-2)^2 - 2 \\
 x = 2 &\text{ で最小値 } -2, \text{ 最大値なし}
 \end{aligned}$$

■150 (4)

$$\begin{aligned}
 y &= -3x^2 + 6x + 2 \\
 &= -3(x-1)^2 + 5 \\
 x = 1 &\text{ で最大値 } 5, \text{ 最小値なし}
 \end{aligned}$$

■151 (1)

$$\begin{aligned}
 y &= x^2 - 4x \\
 &= (x-2)^2 - 4 \quad (0 \leq x \leq 3) \\
 x = 2 &\text{ で最小値 } -4, \quad x = 0 \text{ で最大値 } 0
 \end{aligned}$$

■151 (2)

$$\begin{aligned}
 y &= x^2 - 6x + 7 \\
 &= (x-3)^2 - 2 \quad (0 \leq x \leq 2) \\
 x = 2 &\text{ で最小値 } -1, \quad x = 0 \text{ で最大値 } 7
 \end{aligned}$$

■151 (3)

$$\begin{aligned} y &= -x^2 + 2x + 2 \\ &= -(x-1)^2 + 3 \quad (-1 \leq x \leq 1) \\ x &= 1 \text{ で最大値 } 3, x = -1 \text{ で最小値 } -1 \end{aligned}$$

■151 (4)

$$\begin{aligned} y &= 2x^2 - 8x + 5 \\ &= 2(x-2)^2 - 3 \quad (0 \leq x \leq 4) \\ x &= 2 \text{ で最小値 } -3, x = 0, 4 \text{ で最大値 } 5 \end{aligned}$$

■151 (5)

$$\begin{aligned} y &= -\frac{x^2}{2} - x + 3 \\ &= -\frac{1}{2}(x+1)^2 + \frac{7}{2} \quad (-2 \leq x \leq 2) \\ x &= -1 \text{ で最大値 } \frac{7}{2}, x = 2 \text{ で最小値 } -1 \end{aligned}$$

■151 (6)

$$\begin{aligned} y &= 2(x-1)(x-2) \\ &= 2x^2 - 6x + 4 \\ &= 2\left(x - \frac{3}{2}\right)^2 - \frac{1}{2} \quad (0 \leq x \leq 2) \\ x &= \frac{3}{2} \text{ で最小値 } -\frac{1}{2}, x = 0 \text{ で最大値 } 4 \end{aligned}$$

■152

高さは $8-x$ 、面積を S とすると

$$\begin{aligned} S &= \frac{1}{2}x(8-x) \\ &= -\frac{1}{2}x^2 + 4x \\ &= -\frac{1}{2}(x-4)^2 + 8 \quad (0 < x < 8) \\ x &= 4 \text{ で最大値 } 8 \text{ をとる} \end{aligned}$$

■153 (1)

$$\begin{aligned} 2x^2 + 5x + 1 &= 0 \\ x &= \frac{-5 \pm \sqrt{25-8}}{4} \\ &= \frac{-5 \pm \sqrt{17}}{4} \\ \therefore &2 \text{ 点で交わる} \end{aligned}$$

■153 (2)

$$\begin{aligned} 3x^2 - 7x + 5 &= 0 \text{ の判別式を } D \text{ とすると} \\ D &= 49 - 4 \times 3 \times 5 \\ &= -11 < 0 \\ \therefore &\text{共有点なし} \end{aligned}$$

■153 (3)

$$\begin{aligned} \frac{1}{3}x^2 - 2x + 3 &= 0 \\ x^2 - 6x + 9 &= 0 \\ (x-3)^2 &= 0 \\ x &= 3 \\ \therefore &\text{接する} \end{aligned}$$

■154 (1)

$$\begin{aligned} x^2 + 3x + k &= 0 \text{ の判別式を } D \text{ とすると} \\ D &= 9 - 4k > 0 \\ k &< \frac{9}{4} \end{aligned}$$

■154 (2)

$$\begin{aligned} 4x^2 - 2kx + 9 &= 0 \text{ の判別式を } D \text{ とすると} \\ \frac{D}{4} &= k^2 - 4 \times 9 = 0 \\ k^2 &= 36 \\ k &= \pm 6 \end{aligned}$$

■154 (3)

$$\begin{aligned} kx^2 + x - 1 &= 0 \text{ の判別式を } D \text{ とすると} \\ D &= 1 + 4k < 0 \\ k &< -\frac{1}{4} \end{aligned}$$

■155 (1)

$$\begin{aligned} x^2 + 4x + 3 &\geq 0 \\ (x+3)(x+1) &\geq 0 \\ x &\leq -3, -1 \leq x \end{aligned}$$

■155 (2)

$$\begin{aligned} x^2 - 2x - 1 &< 0 \\ 1 - \sqrt{2} &< x < 1 + \sqrt{2} \end{aligned}$$

■155 (3)

$$\begin{aligned}x^2 - 6x + 9 &> 0 \\(x - 3)^2 &> 0 \\x &\neq 3\end{aligned}$$

■155 (4)

$$\begin{aligned}4x^2 - 12x + 9 &< 0 \\(2x - 3)^2 &< 0 \\&\text{解なし}\end{aligned}$$

■155 (5)

$$\begin{aligned}-x^2 + x - \frac{1}{4} &\leq 0 \\4x^2 - 4x + 1 &\geq 0 \\(2x - 1)^2 &\geq 0 \\&\text{解はすべての実数}\end{aligned}$$

■155 (6)

$$\begin{aligned}\frac{1}{2}x^2 - 4x + 8 &\leq 0 \\x^2 - 8x + 16 &\leq 0 \\(x - 4)^2 &\leq 0 \\x &= 4\end{aligned}$$

■155 (7)

$$\begin{aligned}-4x^2 + 6x - 3 &< 0 \\4x^2 - 6x + 3 &> 0 \\4\left(x - \frac{3}{4}\right)^2 - \frac{9}{4} + 3 &> 0 \\4\left(x - \frac{3}{4}\right)^2 + \frac{3}{4} &> 0 \\&\text{解はすべての実数}\end{aligned}$$

■155 (8)

$$\begin{aligned}2x^2 - 4x + 3 &\leq 0 \\2(x - 1)^2 + 1 &\leq 0 \\&\text{解なし}\end{aligned}$$

CHECK

■156

$$y = \frac{1}{2}(x+2)^2 + 1$$

■157

$$y = x^2 + 3x$$

$$= \left(x + \frac{3}{2}\right)^2 - \frac{9}{4}$$

$$\text{軸: } x = -\frac{3}{2}$$

$$\text{頂点: } \left(-\frac{3}{2}, -\frac{9}{4}\right)$$

■158 (1)

$$y = a(x-2)^2 - 5 \text{ とおける}$$

$$7 = a \times 4 - 5$$

$$a = 3$$

$$y = 3(x-2)^2 - 5$$

■158 (2)

$$y = a(x+1)^2 + q \text{ とおける}$$

$$\begin{cases} 1 = a + q \\ 7 = 4a + q \end{cases} \text{ より } \begin{cases} a = 2 \\ q = -1 \end{cases}$$

$$y = 2(x+1)^2 - 1$$

■158 (3)

$$y = ax^2 + bx + c \text{ とすると}$$

$$\begin{cases} 0 = a + b + c \\ 1 = c \\ 3 = 4a + 2b + c \end{cases} \text{ より } \begin{cases} a = 2 \\ b = -3 \\ c = 1 \end{cases}$$

$$y = 2x^2 - 3x + 1$$

■159 (1)

$$x \rightarrow -x \text{ でおきかえて}$$

$$y = (-x)^2 + 2(-x) - 3$$

$$= x^2 - 2x - 3$$

■159 (2)

$$x \rightarrow -x, y \rightarrow -y \text{ でおきかえて}$$

$$-y = (-x)^2 + 2(-x) - 3$$

$$y = -x^2 + 2x + 3$$

■160

$$y = -x^2 + 6x + 1$$

$$= -(x-3)^2 + 10 \text{ 頂点 } (3, 10)$$

$$y = -x^2 - 2x + 3$$

$$= -(x+1)^2 + 4 \text{ 頂点 } (-1, 4)$$

$\therefore x$ 方向に 4, y 方向に 6 平行移動

■161 (1)

$$y = -x^2 + 4x + 1$$

$$= -(x-2)^2 + 5 \quad (-1 \leq x \leq 1)$$

$$x = -1 \text{ で最小値 } -4,$$

$$x = 1 \text{ で最大値 } 4 \text{ をとる}$$

■161 (2)

$$y = -(x-2)^2 + 5 \quad (1 \leq x \leq 3)$$

$$x = 2 \text{ で最大値 } 5,$$

$$x = 1, 3 \text{ で最小値 } 4 \text{ をとる}$$

■162

x cm の幅で折り曲げるとする ($0 < x < 8$)

断面積を S とすると

$$S = x(16 - 2x)$$

$$= -2x^2 + 16x$$

$$= -2(x-4)^2 + 32$$

$$x = 4 \text{ で最大値 } 32 \text{ をとる}$$

$$\therefore 4 \text{ cm}$$

■163

$x^2 - 6x - k = 0$ の判別式を D とすると

$$\frac{D}{4} = 9 + k$$

(1)

$$D > 0 \text{ すなわち } 9 + k > 0$$

$$k > -9$$

(2)

$$D = 0 \text{ すなわち } 9 + k = 0$$

$$k = -9$$

(3)

$$D < 0 \text{ すなわち } 9 + k < 0$$

$$k < -9$$

■164 (1)

$$2x^2 - 3x - 2 \leq 0$$

$$(2x + 1)(x - 2) \leq 0$$

$$-\frac{1}{2} \leq x \leq 2$$

■164 (2)

$$x^2 + 8x + 16 \geq 0$$

$$(x + 4)^2 \geq 0$$

解はすべての実数

■164 (3)

$$3x^2 + 6x + 4 < 0$$

$$3(x + 1)^2 + 1 < 0$$

解なし

STEP UP

■165

$$y = a(x - p)^2 \text{ とおける}$$

$$\begin{cases} 1 = a(-2 - p)^2 \\ 9 = a(2 - p)^2 \end{cases}$$

$p = 2$ は解でないので、両辺わって

$$\frac{1}{9} = \frac{(2 + p)^2}{(2 - p)^2}$$

$$(2 - p)^2 = 9(2 + p)^2$$

$$p^2 - 4p + 4 = 9p^2 + 36p + 36$$

$$8p^2 + 40p + 32 = 0$$

$$p^2 + 5p + 4 = 0$$

$$(p + 1)(p + 4) = 0$$

$$p = -1, -4$$

$$a = \frac{1}{(p + 2)^2} \text{ より}$$

$$(a, p) = (1, -1),$$

$$\left(\frac{1}{4}, -4\right)$$

$$y = (x + 1)^2$$

$$\text{または } y = \frac{1}{4}(x + 4)^2$$

■166 (1)

$$x^2 + 2(m + 1)x + m = 0$$

の判別式を D とすると

$$\frac{D}{4} = (m + 1)^2 - m$$

$$= m^2 + m + 1$$

$$= \left(m + \frac{1}{2}\right)^2 + \frac{3}{4} > 0$$

$\therefore x$ 軸と 2 点で交わる

■166 (2)

交点の x 座標は

$$x = -(m + 1) \pm \sqrt{m^2 + m + 1}$$

2 交点間の距離は

$$\begin{aligned} & \left(-(m + 1) + \sqrt{m^2 + m + 1} \right) \\ & - \left(-(m + 1) - \sqrt{m^2 + m + 1} \right) \end{aligned}$$

$$= 2\sqrt{\left(m + \frac{1}{2}\right)^2 + \frac{3}{4}}$$

$$m = -\frac{1}{2} \text{ のとき最小となる}$$

■167

$$f(x) = x^2 + 2mx + 1$$

$$= (x + m)^2 - m^2 + 1 \text{ とすると}$$

題意をみたすための条件は

$$\begin{cases} f(-m) = -m^2 + 1 < 0 \\ f(0) = 1 > 0 \\ f(2) = 4 + 4m + 1 > 0 \end{cases}$$

軸 $0 < -m < 2$

$$\Leftrightarrow \begin{cases} m < -1, 1 < m \\ \text{すべての実数} \\ -\frac{5}{4} < m \\ -2 < m < 0 \end{cases}$$

$$\Leftrightarrow -\frac{5}{4} < m < -1$$

■168

$$\begin{aligned}
 y &= 4 - x \geq 0 \\
 x &\geq 0 \text{ とあわせて } 0 \leq x \leq 4 \\
 3x^2 + 2y^2 &= 3x^2 + 2(4 - x)^2 \\
 &= 3x^2 + 32 - 16x + 2x^2 \\
 &= 5x^2 - 16x + 32 \\
 &= 5\left(x - \frac{8}{5}\right)^2 - \frac{64}{5} + 32 \\
 &= 5\left(x - \frac{8}{5}\right)^2 + \frac{96}{5} \quad (0 \leq x \leq 4) \\
 x &= \frac{8}{5} \text{ で最小値 } \frac{96}{5} \text{ をとる。} \\
 \text{このとき } y &= \frac{12}{5} \\
 x &= 4 \text{ で最大値 } 48 \text{ をとる。} \\
 \text{このとき } y &= 0
 \end{aligned}$$

■169

$$\begin{aligned}
 y &= x^2 - 2ax + a^2 - 2 \\
 &= (x - a)^2 - 2 \quad (0 \leq x \leq 2)
 \end{aligned}$$

i) $a < 0$ のとき

$$x = 0 \text{ で最小値 } a^2 - 2 \text{ をとる}$$

ii) $0 \leq a \leq 2$ のとき

$$x = a \text{ で最小値 } -2 \text{ をとる}$$

iii) $2 < a$ のとき

$$x = 2 \text{ で最小値 } a^2 - 4a + 2 \text{ をとる}$$

■170

長方形の縦を $2y$ 、横を $2x$ とすると

$$x^2 + y^2 = 3^2 \text{ が成り立つ}$$

$$\therefore y^2 = 9 - x^2$$

$$y^2 \geq 0 \text{ なので } x^2 \leq 9$$

$$\therefore -3 \leq x \leq 3$$

面積 S は

$$S = 2x \cdot 2y$$

$$= 4xy$$

$$= 4x\sqrt{9 - x^2}$$

$$= 4\sqrt{-x^4 + 9x^2}$$

$$= 4\sqrt{-\left(x^2 - \frac{9}{2}\right)^2 + \frac{81}{4}} \quad (0 \leq x^2 \leq 9)$$

$$x^2 = \frac{9}{2} \text{ のとき面積が最大となる。}$$

$$\text{このとき } y^2 = \frac{9}{2}$$

$$x = y \text{ なので正方形}$$